

Fusion360 Making Components

This document contains step-by-step Instructions to define a component with Multiple ground pins in Fusion 360 electronics.

Steps

1. Open or Create Your Library: Launch Fusion 360 Electronics and open the library where you want to define the component. If needed, create a new library or edit an existing one.
2. Create or Edit the Schematic Symbol:
 - In the Library Editor, select or create the schematic symbol for your component.
 - Add a single GND pin to the symbol (or use an existing one if it's already there). Name it "GND" (or similar). Do not create multiple separate GND pins unless the grounds are functionally different (e.g., analog vs. digital).
3. Create or Edit the Footprint:
 - Switch to the footprint (package) editor in the Library.
 - Define all the physical pads that correspond to ground connections on the IC (e.g., pads 5, 8, 17, 20, and any exposed pad like EP). Ensure these pads are correctly positioned and sized based on the datasheet.
4. Link the Symbol and Footprint in the Device Editor:
 - In the Library Editor, go to the Device section (this is where you combine the symbol and footprint).
 - Add your schematic symbol and footprint to the device if they're not already linked.
 - Click on the "Connect" button (or right-click and select "Connect") to open the pin-to-pad mapping dialog.
5. Map the Single GND Pin to Multiple Pads:
 - In the Connect dialog, select the single "GND" pin from the schematic symbol on the left side.
 - On the right side, hold Ctrl (or Cmd on Mac) and multi-select all the relevant ground pads from the footprint (e.g., pads 5, 8, 17, 20, EP).
 - Click "Connect" to assign the single GND pin to all selected pads. This groups them electrically as the same net.
6. Verify and Save the Device:
 - Close the Connect dialog and check the mappings: The GND pin should now show connections to multiple pads.
 - Save the library and update any open schematics/PCBs if prompted.
7. Test in Schematic and PCB:
 - Place the component in a new or existing schematic.
 - Connect a single GND net to the component's GND pin.

- Push the schematic to the PCB layout: All ground pads should appear as part of the same GND net, with no extra air-wires between them.
- Route traces or pours to the pads as needed; the software will treat them as interconnected.

This setup ensures that in the schematic, you only manage one GND connection, while the PCB handles routing to all physical ground pads efficiently. If your IC has non-interchangeable grounds, repeat the process with separate pin names (e.g., AGND and DGND) instead of grouping them.

Autodesk's recommended approach

These notes came from Fusion 360 / EAGLE forums and documentation. For typical cases where multiple ground pins on an IC are internally connected and interchangeable is to use a single schematic pin mapped to all corresponding footprint pads via the Connect dialog—this avoids unnecessary clutter in the schematic and ensures proper net connectivity in the PCB without manual fixes.

Pin names in a schematic symbol **must be unique** within the symbol (the software enforces this rule to prevent ambiguity in net assignment). If you need multiple GND pins visible in the symbol (e.g., for clarity or when they are not fully interchangeable), the standard workaround is to name them GND@1, GND@2, GND@3, etc.—the "@" suffix makes each name technically unique while collapsing them to a single "GND" net in the schematic and PCB views. However, this can lead to complications: it forces you to place and connect multiple GND pins individually in the schematic (increasing clutter, especially on large designs), risks accidental separation into different nets if not handled carefully, and may confuse autorouting or ERC/DRC checks if the @ syntax is misunderstood or inconsistently applied.

For best practice on most SMT ICs with equivalent grounds, stick to the single-pin mapping method unless the data-sheet explicitly requires separate treatment.